Completing the Application Form Guidance

1. Application Summary Information

Host Organisation

University of Kent

Application Type
NIHR Academy Strategic Theme
Social Care

Partnership fellowship
NA

Advanced Fellowship application type
Dementia (NIHR Highlight notice)

Research Title – 300 word limit

Healthy you – keeping yourself healthy while supporting someone living with dementia: development and feasibility testing of a complex intervention

Research Type

- [ ] Clinical trial (phase 2)
- [ ] Clinical trial (phase 3a – efficacy)
- [ ] Clinical trial (phase 3b - effectiveness)
- [ ] Clinical trial (phase 4)
- [x] Feasibility study
- [ ] Pilot study
- [ ] Secondary data analysis
- [ ] Cohort study
- [ ] Epidemiological
- [ ] Meta analysis
- [x] Qualitative study
- [ ] Retrospective review
- [x] Survey
- [ ] Evidence Synthesis

- [ ] Evaluation
- [ ] Other (please specify)

Proposed Start Date

1st January 2025

Do you wish to hold an Advanced Fellowship for the full time equivalent of 2, 3, 4 or 5 years?

4

WTE of Award

0.90

Duration

53 months

2. Applicant CV

It is recognised that the outputs in terms of grants and publications will vary depending on your experience, profession and level of award being applied for. Linking to and using the response to the *Applicant Research Background* section you should highlight the impact of outputs listed under research grants and make the case for why your experience and outputs make you suitable for the fellowship being applied for.

Publication Record

Relevant Prizes, Awards and other Academic Distinctions

ORCiD

3. Applicant Research Background

Professional background

Not a registered health professional

Profession (Not a Registered Health Professional)

Psychologist

Please describe your research career to date – 1000 word limit

I am psychologist who works in adult social care research with particular interest in family members who support someone living with dementia (unpaid carers). There are two key messages from unpaid carers that inspired this Fellowship: 'carers are never asked what they needed/what they want' and 'you can't pour of an empty jug. Carers are out of steam.' As such, I see this NIHR Advanced Fellowship as a beginning of my long-term commitment to help unpaid carers to have a good mental and physical wellbeing.

I started to be interested in a topic of wellbeing and health of unpaid carers who support someone living with dementia while working on projects that focused on understanding of unmet social care needs of people living with dementia and their unpaid carers (2019 – to present). This has given me an incredible opportunity to build relationships with carers, health and adult social care professionals who support them and representatives from the voluntary sector across England. It has also enabled me to identify that there is a need for research that would specifically focus on physical and mental wellbeing of unpaid carers who support someone living with dementia. As a result of this unmet research need I have successfully applied for NIHR Three Schools: Dementia Research Programme Award (£32,166; Personal Career Award; July 2022-January 2023). This award was crucial in co-producing this Fellowship application as it allowed me to build a research team with eight unpaid carers of people living with dementia (co-researchers). Between January and June 2023 we met online every month for two hours and worked together on what the focus of the Fellowship should be. In the second half of the year, we continued to meet monthly to discuss all work packages proposed in this Fellowship, dissemination activities and their role in the Fellowship if awarded. All co-researchers wish to continue to be part of the research team as they are all committed to improving lives of other carers; value the opportunity to learn from each other and value the emotional support they can get from each other. Alongside this Fellowship application, we also co-produced a blog[1] and we engaged with an illustrator who captured our work on how unpaid carers of people living with dementia can meet some of their goals when it comes to their own health and wellbeing[2]. In addition, our team also applied together with local artist for SHAPE Involve and Engage grant from the British Academy (£8000, submitted in June 2023, unsuccessful) to enable us to hold a mixed-media workshop with unpaid carers of people living with dementia followed by an exhibition to increase awareness of what it means to be a carer for a person with dementia. We are also working together on academic paper and a podcast episode (for Dementia Researcher) to reflect on our experience of co-producing this Fellowship application (both outputs suggested by the co-researchers themselves).

Importantly, the NIHR Advanced Fellowship not only represents an incredible opportunity to start to build my career around the topic I am passionate about but also to give me an opportunity to accelerate my own research independence through leading on a substantive

research project. While this is going to be first time I am going to lead on a substantive research project in an area that is relatively new to me, I am confident I will be able to deliver suggested research with the support of my mentors, co-researchers, external partners and support teams across University of Kent. The training element of the Fellowship is particularly important in gaining relevant skills in becoming a researcher who has an influence beyond academia. Specifically, I would like to make the needs of unpaid carers visible to present and future health and social care professionals as well as policy makers. Therefore, I would like to gain skills in leadership; advocacy; policy influence; and use of art media to achieve this goal. Naturally I recognise that while learning on 'the go' is crucial source of experience, I have learned that the highest quality research can be only conducted with strong knowledge and skills base. As such I also plan to attend specific training to deepen my methodological skills and I will get mentorship from Brighton & Sussex Clinical Trials Unit in three areas that we identified together: data management, study management and statistical advice.

The suggested Fellowship builds on my research experience so far. I have acquired experience in all stages of the research process – from protocol development, ethics, conduct of studies of different designs (both quantitative and qualitative) through to dissemination of the results. My early postdoctoral career focused on developing and evaluating behavioural change interventions. For example, as a Career Development Fellow at University of Cambridge (2014-2016), I tried to understand the impact of communicating risk estimates on physicians' and patients' behaviour, well-being and other relevant clinical outcomes. I expanded what I learned from cardiovascular disease prevention on dementia prevention during my time as a Research Fellow at University of Exeter (2017-2019). Since 2019, I have been involved in a range of projects in adult social care. My expertise in adult social care has been recognised by the NIHR and I was invited to review research applications submitted to NIHR Research (Programme) for Social Care funding calls. In addition, as a core member of the NIHR Research Design Service South East team (research advisor on adult social care (10% FTE, from January 2021), I was invited to be part of NIHR Research Support Service call application (approx. £12,000,000). I also provided recommendations to Department of Health and Social Care which tool measuring work-related quality of life to include as part of Adult social care workforce survey. I am also a member of PhD supervisory team (from September 2023) that focuses on experiences of carers who do not live in the same household as a person with dementia. Importantly, I started to lead on applications that combine my background in psychology and my experience in adult social care research.

Has this application been previously submitted to this or any other funding body? – 500 word limit

No

Mitigating Factors - 500 word limit

There are two key mitigating factors that impacted my research career so far: 1. Experience of severe depression and anxiety and 2. Covid-19 pandemic.

I was diagnosed with severe depression and anxiety in March 2019 while working as a Research Fellow at University of Exeter. Following the diagnosis, I took an extended period of sick leave (three months; maximum allowed for the full pay allowance) followed by phased return to work. As a result of being unwell, I decided that I wanted to live closer to London to be able to visit regularly my family and friends who live abroad. Importantly, I wanted to take a step back from the research I did until this point. Subsequently, I applied for a position at University of Kent and shifted the focus from health to adult social care research.

I joined University of Kent in September 2019. Six months later the first lockdown was enforced on 26th March 2020. This was followed by series of lockdowns and restrictions until December 2021.

Since joining University of Kent, I have been working across several projects that include primary data collection in adult social care. While data collection in adult social care is challenging and time demanding, this became a particular issue during the COVID-19 pandemic. More specifically, adult social care was one of the most impacted sectors during this time. This meant for example that:

1. Data collection was completely halted and delayed. This was due to, e.g. being unable to obtain research governance approvals. Many local authorities temporarily suspended their research governance approval process. Importantly local authorities and adult social care organisations were unable to support recruitment of participants.

2. Changes to data collection. As we experienced issues with data collection, we had to explore new recruitment pathways: e.g. data collection through NHS instead of adult social care organisations.

3. Data collection was time consuming: for example, to meet our target of recruiting 300 participants, I had to send 10 000 individual emails to potentially eligible participants - this consumed majority of my hours dedicated to the project. In addition, I was responsible to prepare study packages and send them to participants.

The above are few of the examples, how the Covid-19 pandemic impacted projects I was involved in. The key impact was that I spend all my time on ensuring we would collect data on time and meet our recruitment targets. Hence, I had no time to dedicate to writing either research papers or applying for funding applications. As the COVID-19 pandemic started at the time I was still recovering from episode of mental illness, it was vital to me to prevent any relapse (e.g. by not taking up extra work besides successful delivery of the projects in time and on target). Since 2022, I have been able to resume applying for research grants, leading on publications and leadership positions within my organisations on top of my day to day tasks.

Plain English Summary of Research (600 words)

Aim(s) of the research: The focus of this Fellowship is on family members who support someone living with dementia. We would like to understand how we can support them to look after their own health and wellbeing. Therefore as a first step, we will develop an intervention that will help family members of someone with dementia to achieve their health and wellbeing goals. For example they may want to go for a walk more often or improve their time management. Once we develop this intervention, we will test whether it is practical, e.g. what are the strengths and weaknesses of the proposed intervention. We will also test whether the intervention is acceptable for unpaid carers and for those who will be delivering it.

Background to the research: Across the UK there are 700,000 family members and friends of people living with dementia. In literature they are often referred to as carers, unpaid carers, family carers or supporters etc. but the family members and friends may not use any of these terms to describe themselves. They often help the person with dementia in a range of ways, including with personal care, providing emotional support and arranging support from health and adult social care services. Unpaid carers themselves highlighted that 'It is exhausting to be a carer – it takes so much out of you and you do not have a support'. Currently, we know very little about how we can support unpaid carers to achieve their health and wellbeing goals.

Design and methods used: Changing behaviour, for example walking every day for 30 minutes, is not an easy task. It is even more challenging to make any changes in behaviours when someone is an unpaid carer. In this Fellowship we will follow step by step guide called the 'Behaviour Change Wheel Model' to understand how we can support unpaid carers to reach their health and wellbeing goals. In Stage 1 we will conduct needs assessment (through reviewing existing evidence and collecting information among unpaid carers) to identify the problem, e.g. should we focus on helping carers to increase their physical activity or help them to cope with terminal illness. In Stage 2 we will use a range of methods e.g. survey, individual interviews, and group discussions to design the intervention itself. In Stage 3 we will test whether the proposed intervention is practical and acceptable by the unpaid carers and the people who will deliver it.

Patient/service user, carer and public involvement: Unpaid carers played a central role in co-producing this Fellowship application. In co-production, researchers, practitioners and the public work together from the start to the end of the project and the research is jointly owned. Over the period of twelve months we met together on monthly basis to decide together the focus of the Fellowship, how we are going to answer our research questions, disseminate results etc. We were a team of ten people - eight people self-identified as former or current unpaid carer living across England and Scotland. They will continue to work on the Fellowship as co-researchers – this means they will be part of the core research team, they will own the research and be responsible for it.

Dissemination: Unpaid carers highlighted that it is important that the results from this Fellowship reach wide range of people including carers, health and social care professionals and shared throughout the whole project regularly. We will use range of approaches, for example: Podcast/YouTube channel; webinars; articles for Alzheimer's & Dementia Journal or the Journal of Dementia Care; working with illustrator to capture some of the findings etc.

Scientific abstract 500 words

Research question: Stage 1: a) What are the health and wellbeing goals of unpaid carers of people living with dementia and what is their association with their overall health and wellbeing? Stage 2: a) Which behaviours need to be changed to allow unpaid cares to meet their wellbeing goals?; b) What it will take to bring about the desired change (e.g. capability, opportunity, motivation)?; c) What types of intervention approaches might help?; d) What specific components should the intervention involve? Stage 3: Is the proposed intervention feasible and acceptable for unpaid carers of people living with dementia?

Background: Across the UK there are 700,000 family members and friends of people living with dementia (unpaid carers). While being a carer can be rewarding, caring is associated with a range of mental and/or physical health difficulties such as feeling anxious, lonely, depressed, having difficulties to sleep and experiencing back problems. The Department of Health and Social Care recently announced that to support health and wellbeing of unpaid carers it allocated £327 million to allow them to take time from caring. While there is evidence suggesting that unpaid carers have worse physical and health outcomes, we know very little about what their health and wellbeing goals are and how we can support them to achieve their health and wellbeing goals.

Aims and objectives: The ultimate goal of this Fellowship is to design and evaluate feasibility of behaviour change intervention that will support unpaid carers of people living with dementia to look after their own physical and mental health and wellbeing.

Methods: The study will use mixed methods across three stages of work. The proposed work is informed by Medical Research Council guidance for developing and evaluating complex interventions; Behaviour Change Wheel model; and Co-production. Behaviour change wheel framework provides a systematic and comprehensive process for designing behaviour change interventions. Eight unpaid carers co-produced the suggested research and will be part of core research team throughout the project. In Stage 1 (needs assessment) we will conduct a systematic review and survey to identify health and wellbeing goals of unpaid carers of people living with dementia. In Stage 2 (intervention development) we will use range of methods to define target behaviours, to identify what needs to change in the person and/or the environment in order to achieve the desired change in behaviour, what specific techniques and components the intervention should involve. In Stage 3 we will conduct a feasibility study to test whether the proposed intervention is feasible and acceptable for unpaid carers.

Timelines for delivery: Stage 1: 1-12 months; Stage 2: 13-43 moths; Stage 3: 43-53 months.

Anticipated impact and dissemination: A functional, feasible, and acceptable intervention to enable unpaid carers of people living with dementia to meet their health and wellbeing goals. For dissemination of results, we will use range of approaches, for example: Podcast/ YouTube channel; webinars; articles for Alzheimer's & Dementia Journal or the Journal of Dementia Care; working with illustrator to capture some of the findings etc.

Detailed Research Plan – 5000 word limit

What is the problem being addressed?

Why is this research important in terms of improving the health and/or wellbeing of the public and/or to patients/service users, carers and health and care services?

Review of existing evidence: how does the existing literature support this proposal?

What is the research question/aims and objectives?

The ultimate goal of this Fellowship is to design and evaluate feasibility of behaviour change intervention that will support unpaid carers of people living with dementia to look after their own physical and mental health and wellbeing. Throughout the Fellowship I will answer following questions:

Stage 1:
a) What are the health and wellbeing goals of unpaid carers of people living with dementia?
b) What is the association between Exercise; Nutrition; Coping with stress; Coping with terminal illness; Coping with emotions; Getting help; Knowledge; Time management and overall physical and mental health among unpaid carers of people living with dementia? and Are there effective interventions to improve: Exercise; Nutrition; Coping with stress; Coping with terminal illness; Coping with emotions; Getting help; Knowledge; Time management among unpaid carers of people living with dementia?

Stage 2: a) Which behaviours need to be changed to allow unpaid cares to meet their wellbeing goals?; b) What it will take to bring about the desired change (e.g. capability, opportunity, motivation)?; c) What types of intervention approaches might help?; d) What specific components should the intervention involve?

Stage 3: Is the proposed intervention feasible and acceptable for unpaid carers of people living with dementia?

Project Plan

The study will use mixed methods across three stages of work. The proposed work is informed by Medical Research Council guidance for developing and evaluating complex

interventions[3]; Behaviour Change Wheel model[4]; and Co-production [5]. In summary, in Stage 1 (needs assessment; 1-12 months) we will conduct a systematic review and survey to identify health and wellbeing goals of unpaid carers of people living with dementia. In Stage 2 (intervention development; 13-43 moths) we will use range of methods to define target behaviours, to identify what needs to change in the person and/or the environment in order to achieve the desired change in behaviour, what specific techniques and components the intervention should involve. In Stage 3 (43-53 months) we will conduct a feasibility study to test whether the proposed intervention is feasible and acceptable for unpaid carers.

Medical Research Council guidance for developing and evaluating complex interventions[3] divides complex intervention research into four phases: development or identification of the intervention, feasibility, evaluation, and implementation. This Fellowship focuses on first two phases: development of the intervention and feasibility study. The development of the intervention itself is informed by Behaviour Change Wheel model[4]. It is a step-by-step guide using COM-B (capability-opportunity-motivation-behaviour) model. There are other approaches to developing interventions e. g. MINDSPACE[6], Intervention Mapping[7] but the Behaviour Change Wheel model is the most comprehensive approach to behaviour change interventions (developed from 19 frameworks of behaviour change model[8]). As highlighted by the authors[4], the key benefit of using the Behaviour Change Wheel model is that '*it encourages intervention designers to consider the full range of options and choose those that are most promising through a systematic evaluation of theory and evidence*.'

Behaviour Change Wheel model includes following steps[4]: a) Understanding the behaviour: specifying the behaviour that is the target of the intervention, b) followed by and introduction to a simple method of analysing what needs to change first using COM-B model: is greater capability, more opportunity or stringer motivation required?; c) Identifying intervention options: how best to achieve change, e.g. by education, persuasion, training or changing this environment, before considering how policy such as fiscal policy or service provision might facilitate the process, and d) Identifying content and implementation options: a wide range of techniques that are the active ingredients of every behaviour change intervention. Decisions are made about how to deliver intervention: e.g. when and how often; is it best face to face, by internet or in some other way.

Importantly suggested project plan has been co-produced with eight unpaid carers of people living with dementia (co-researchers). In co-production, researchers, practitioners and the public work together from the start to the end of the project[9]. One of the key principles in co-production is that '*the research is jointly owned and people work together to achieve a joint understanding*'[9]. It has been recognised that involving members of the public in designing and conducting research can ensure that the project plan best addresses the needs of the target population[3]. More details on how co-researchers contributed to this application and how they will continue to support delivery of the project plan is described in section about Patient & Public Involvement.

Stage 1: Needs assessment; 1-12 months: survey and systematic review
As highlighted earlier, unpaid carers of people living with dementia experience range of mental and physical health issues (e.g.[10-17]). Between January and June 2023 eight unpaid former or current carers (co-researchers) of people living with dementia who live across Scotland and England identified 22 health and wellbeing concerns experienced by carers of people living with dementia. These included: stress-chronic, stress-situational; frustration/ anger; nutrition (e.g. quick fixes, binging, snacking), lack of exercise, lack of sleep, loneliness, lack of feelings of connection, isolation, burned out, lack of connection to communities with similar backgrounds, dental care, getting own medication, type 2 diabetes, high blood pressure, feeling guilt, mental well-being, exhaustion, grief, feelings of shame, stigma and prejudice/discrimination. They then narrowed these down to five key health and wellbeing concerns for carers of people living with dementia: 1. Mental health and wellbeing (stress, frustration, anger, burned out, guilt, grief including anticipatory grief); 2. Self-care (nutrition, being able to exercise, lack of sleep); 3. Connection with people and communities (loneliness, isolation); 4. Physical wellbeing (feeling exhausted, dental care, getting own medication); and 5. Feeling of stigma and shame. As a last step, co-researchers suggested eight goals they would like to work towards when it comes to their own health and wellbeing: 1. Exercise, 2. Nutrition, 3. Coping with stress, 4. Coping with terminal illness, 5. Coping with emotions, 6. Getting help, 7. Knowledge and 8. Time management[1].
Although these suggestions by co-researchers offer invaluable insight to the perspectives of unpaid carers of people living with dementia, we cannot generalise their views on the whole population of the unpaid carers. As highlighted by co-researchers: '*Everyone is different and it is important to take time to evaluate people's needs*' and '*The key things to be addressed are many, everyone has a voice and different experience. Such is the diversity of a problem that is not being tackled sufficiently well enough.'*

Importantly, the fact that the views of other unpaid carers can differ from co-researchers involved in this application was highlighted by the results from the survey that one of the co-researchers conducted in her carers' group. To be more specific she shared with nine carers eight goals that the co-researchers identified and asked them whether these goals resonate with them and to order them from most important to least important. While all the goals resonated with them and none new goals were identified, the importance of the goals differed.

Therefore as a first step I will conduct **a survey** that will aim to identify the health and wellbeing goals of unpaid carers of people living with dementia.

Stage 1a: Needs assessment; 1-12 months: survey
Aim: To identify top three health and wellbeing goals of unpaid carers of people living with dementia.

Study design: a self-administered questionnaire for unpaid carers in England. Participants will be able to choose between a postal or online version (using Qualtrics XM Platform™).

Being able to choose between postal and online version was suggested by co-researchers. Having a paper version available was also highlighted as important to me by research staff across NHS sites in other research that involved unpaid carers of people living with dementia[18].

Survey: all study documents including survey will be developed together with eight co-researchers. The survey will include background questions about the unpaid carer: caregiving situation: former vs current carer, whether the carer and person with dementia lives in the same household, relationship to a person living with dementia, estimated hours of care per week, whether the carer provides help with personal care or medicines and any additional support received from other family members and friends, health care and adult social care services. Sociodemographic characteristics: age; gender; ethnicity; region; urban vs rural area; socioeconomic status; English as a second language. Impact of caring on health as used in the Survey of Carers in Households – England 2009/10[19]: carers will be asked whether their own health had been affected by the care they provide and could choose from different options (e.g. feeling tired; feeling depressed; loss of appetite; disturbed sleep; physical strain etc.). We will also ask about any mental and physical health issues experienced by carers. Mode of data collection: paper survey or online.

The main section will ask participants to choose top three health and wellbeing goals for unpaid carers of people living with dementia. They will be able to choose from the eight goals suggested by co-researchers as described earlier. Participants will have an opportunity to suggest other health and wellbeing goals and will be provided also with free text box to explain their decision.

Participants and setting: unpaid carers of people living with dementia across England. Both former (within five years[20]) and current carers will be eligible to take part, if they provid(ed) unpaid support to a relative, partner/spouse or friend with dementia. They will have to be at least 18 years old and able to provide written consent. While this survey is not James Lind Alliance Priority Setting Partnership per se, I used their recommendations on sample size to determine how many participants need to be recruited for this survey. At present, there is no recommended maximum or minimum number of responses. Survey responses from previous priority settings have varied from 100s to 1,000s[21]. As such there will be no predetermined sample size but instead we will focus on distributing the survey through multiple channels. We will recruit participants through GP practices, memory services, pharmacies, social media (e.g., Twitter), previous studies, online dementia research websites (e.g. Join Dementia Research (JDR)), carer organisations and dementia services. The recruitment will be opened up to four months.

Analysis: descriptive analysis (e.g. frequency, median). We will also explore whether there are any differences in goals between e.g. former and current carers; carers from different socioeconomic backgrounds etc.

Equality, diversity and inclusion (EDI): we will ensure that we will recruit participants from range of backgrounds including recruiting participants from specific areas to sample different ethnicities, socioeconomic backgrounds and levels of access to health and dementia care services. We will also engage with organisations that specifically support LGBTQ+ communities. I am also a member of Dementia Community EDI specialty interest group and will draw on the members' expertise. Examples of the organisations that we will engage with are: Opening Doors; Switchboard Dementia; LGBTQ+ Dementia Advisory Group team; The Ethnic Minorities in Canterbury (EMIC); Diversity Resource International; the West Sussex Alzheimer's Society (link established with chair of the Dementia Strategy Minorities Sub-group) and Carers Support West Sussex (link with Bi-lingual Dementia Navigator). We will also provide translated documents in multiple languages.

Outcome: top three health and wellbeing goals of unpaid carers of people living with dementia. In addition we will establish the suitability of recruitment pathways and whether any changes to recruitment has to be introduced for the Stage 2 of the proposed Project Plan.

Dissemination: The results will be shared with participants and participating health care and social care organisations. We will also disseminate the results through the Steering Committee and co-researchers will use their established links, e.g. Mobilise, Facebook groups, their own carers' groups, Dementia Friendly East Lothian, Local Advocacy groups etc. Co-researchers suggested following outputs to disseminate the findings: blog; leaflets; webinar; article for Alzheimer's Society journal or the UK Dementia Journal. We will also write peer-reviewed paper and co-present the results at a conference.

Stage 1b: Needs assessment; 1-12 months: systematic review and meta-analysis

Alongside the survey (described in Stage 1a), we will conduct systematic review and meta-analysis.

Research questions:

1. What is the association between Exercise; Nutrition; Coping with stress; Coping with terminal illness; Coping with emotions; Getting help; Knowledge; Time management and overall physical and mental health among unpaid carers of people living with dementia?
2. Are there effective interventions to improve: Exercise; Nutrition; Coping with stress; Coping with terminal illness; Coping with emotions; Getting help; Knowledge; Time management among unpaid carers of people living with dementia?

Methods: The protocol for this review will be registered a priori in the PROSPERO database. The reporting will follow the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guideline[22].

Eligibility criteria, information sources and search strategy: only studies published in English in peer-reviewed journal will be included. Similarly as in another review focusing on unpaid carers we will limit search by publication date (last 10 years)[23]. We will include

observational studies to answer research question 1 and randomised trials to answer research question 2. We will only include studies that included unpaid carers of people living with dementia. The outcome for research question 1 is physical or mental health-related outcomes of unpaid carers, including health-related quality of life. The outcome for research question 2 is Exercise; Nutrition; Coping with stress; Coping with terminal illness; Coping with emotions; Getting help; Knowledge; Time management in unpaid carers. We will search electronic databases, ongoing trials registers and Google Scholar. In addition, we will perform a manual search of the reference lists of included studies. Search strategy will be developed in consultation with university librarian. The titles, abstracts and full texts will be screened by at least two reviewers.

Risk of bias: We will use Critical Appraisal Checklists[24] to assess the risk of bias in included studies.

Data synthesis and statistical analysis: Detailed statistical plan will be reviewed by statistician. For the observational studies, I will conducted a random effects meta-analysis of cross-sectional findings following the procedure outlined by Borenstein et al.[25] similarly as in other review I was involved in[26]. For the randomised controlled trials, where possible I will combine data from different studies using random effects meta-analysis.

EDI: Information will be extracted and results will be summarised based on unpaid carer's ethnicity/race, age, gender and sexual orientation.

Outcome: Associations between eight health and wellbeing goals and overall physical and mental health among unpaid carers of people living with dementia will be identified. Comprehensive list of interventions including whether they work, in which setting they were tested and with whom.

Dissemination: Similar dissemination strategies will be used as suggested in Stage 1a. The key planned outputs from this stage are: infographic, conference, peer-reviewed paper.

At the end of Stage 1 (Needs assessment) I will organise the Steering committee (SC) meeting. The SC will include co-researchers, mentors and academics with relevant expertise (including intervention developer, statistician and health economist), health and social care professionals, and representatives from independent charities. The aim of this SC meeting will be to review the findings from Stage 1 and decide together which three goals will be taken forward to Stage 2. As suggested by Medical Research Council guidance for developing and evaluating complex interventions[3] we will consider following criteria when making this decision: priority; opportunity to intervene; ability to change; size of potential benefit; stakeholder desirability for an intervention etc.

Stage 2: **Intervention development; 13-36 moths**

Stage 2 will focus on development of intervention for three health and wellbeing goals as identified by the end of Stage 1. The Intervention development will be informed by Behaviour Change Wheel model[4]. It includes following steps: a) Understanding the behaviour; b) Understanding what needs to change; c) Identifying intervention options; and d) Identifying content and implementation options.

Stage 2a: **Understanding the behaviour; 13-24 moths: Delphi consensus exercise and focus groups**

Aim: to specify the behaviour that is the target of the intervention

Study design: first we will conduct Delphi consensus exercise with a series of rounds where participants will be asked: What behaviour needs to change? Who is involved? What makes it more or less likely to succeed? Then we will use focus groups to decide which behaviour to start with using the criteria of impact, likelihood of change, positive or negative spill-over to other behaviours and ease of measurement of outcomes. All study documents including the content of the Delphi consensus exercise and topic guides will be developed together with eight co-researchers.

Delphi consensus exercise: The Delphi consensus exercise is designed to elicit a 'consensus of opinion' from a group of experts. Participants remain anonymous and provide feedback as part of the iterative process[27, 28]. The aim of the Delphi consensus exercise will be to develop a consensus among unpaid carers of people living with dementia on a list of target behaviours and actors. It will include background questions about the unpaid carer and their sociodemographic characteristics (similar to Stage 1a). The initial list of 'What behaviour needs to change?' **'**Who is involved?' **'**What makes it more or less likely to succeed?' for the expert panel to consider, will be based on suggestions from eight co-researchers involved in this Application. Examples for e.g. nutrition can be found here[1]. We will conduct 3-round online modified Delphi consensus exercise using ExpertLens™,[29] an online platform and methodology for conducting modified Delphi panels. It offers a tool that offers a data collection platform with survey, automated data analysis, and discussion functionalities.

Focus groups: designers can select more than one target behaviour though it is advised to limit the intervention to just one or a few behaviours in the first instance[4]. Building on the results from the Delphi consensus exercise we will conduct up to five focus groups to decide which behaviours to start with using the criteria of impact, likelihood of change, positive or negative spill-over to other behaviours and ease of measurement of outcomes. The co-researchers agreed that that focus groups are a good way how to answer our research questions. It was highlighted that '*in groups people have a tendency to open up and relate their experiences'*

Participants and settings:
<u>Delphi exercise:</u> unpaid carers of people living with dementia across England. Both former (within five years[20]) and current carers will be eligible to take part, if they provid(ed)

unpaid support to a relative, partner/spouse or friend with dementia. They will have to be at least 18 years old and able to provide written consent. It is important to note that there is no agreement on the minimum panel size required for content validity in Delphi studies. Recommendations vary from 5-30 participants [30-32]. Given the diversity of experiences in carers of people living with dementia, we will aim to recruit up to 30 participants for the Delphi consensus exercise. We will include and build on our recruitment strategies as suggested in Stage 1a. Participants taking part in Stage 1a will be also asked whether they would agree to be contacted in future about research studies. The recruitment and data collection will be opened up to four months.

Focus groups: range of stakeholders (actors/who is involved) living across England as identified in Delphi consensus exercise. At the moment with co-researchers we identified following stakeholder groups: unpaid carers; other family members and friends; health care professionals; people working in adult social care and social work. However, '*it is important that everyone in the focus group understands what an unpaid carer does/the lived experience needs to be right at the forefront - not even having time to brush teeth*.' It was highlighted that the focus groups should be small, e.g. 3-5 people; the groups should be homogenous rather than mix of perspectives but the key messages from one group should be shared with others. Both online and in-person options will be provided to conduct the focus groups. We will build on recruitment strategies as established in previous Stages. It was suggested that it would be good to hold some of these focus groups e.g. in a local day centre where there are professionals already there and the carer and the person who has dementia all come together, have a cup of tea, a sandwich etc. To avoid the burden on health care and social care professionals we will try to organise the focus groups e.g. as part of their team meetings for example.

EDI: similar considerations will apply as in Stage 1a. The focus groups will be organised across England, the length of each focus group will be up to one hour and times will be provided outside typical working hours. We will contribute to carers replacement costs. If people do not wish to participate in a focus group e.g. due to language barrier an option will be provided to have an individual interview with an interpreter present. We acknowledge that those who do not have access to internet/do not wish to engage in Delphi online exercise will be excluded from this study.

Analysis:

Delphi exercise: Rating data are analysed automatically using the RAND/UCLA Appropriateness Method (RAM) to determin group consensus[33]. By combining numeric answers with discussion comments, ExpertLens not only identifies areas of agreement in large and diverse groups, but also allows to understand what participants disagree on and why their perspectives vary[29].

Focus groups: The focus groups will be audio-recorded using encrypted digital voice recorders and transcribed. The content of the focus groups will be analysed using content analysis.

Outcome:
Delphi exercise: for each health and wellbeing goal core behaviours will be identified. We will understand: What behaviour needs to change?' 'Who is involved?' 'What makes it more or less likely to succeed?'
Focus groups: the target behaviour that we wish to change is specified

Dissemination: Similar dissemination strategies will be used as suggested in Stage 1. The key planned outputs from this stage are: lay summaries, conference, peer-reviewed paper. In addition we will present results from Delphi consensus exercise and focus groups at Kent Research Partnership Community of Practice Group (complex needs) to see whether our findings resonates with their experiences. This group include mix of people with lived experience as well as professionals who support them.

Once the Delphi exercise and focus groups are finalised we will organise the Steering Committee meeting to share the results with the group. We will make then final decision on which target behaviours the intervention should address. We will ensure that this decision is culturally, socially and context driven and acceptable. The target behaviours will be described in terms who needs to perform the behaviour, what the individuals need to do differently, when, where, how, and with whom will they do it[4].

Stage 2b: Specify what needs to change; 25 – 36 moths: focus groups
The COM-B model suggests that changing the incidence of any behaviour of an individual, group or population involves changing one or more of the following: capability (physical/ psychological), opportunity (physical/social), and motivation (reflective/automatic) relating either to the behaviour itself or behaviours that compete with or support it[4]. These high-level constructs can be understood further using the Theoretical Domains Framework (TDF)[34]. The more accurate this analysis of the target behaviour, the more likely it is that the intervention will change the behaviour in the desired direction[4].

Aim: to identify what needs to change in the person and/or the environment in order to achieve the desired change in behaviour.

Study design: we will use at least five focus groups to explore the behaviours as specified at the end of Stage 2b. Again the co-researchers agreed that that focus groups are a good way how to answer our research questions. It was highlighted that the focus groups should be small, e.g. 3-5 people. Both online and in-person options will be provided to conduct the focus groups. We will build on recruitment strategies as established in previous Stages. Questions asked as part of the focus groups will be informed by both COM-B model[4] and

TDF[34]. We will also ask what could help participants overcome challenges/barriers in changing their behaviour mentioned during the focus groups. In addition, we will also collect background information (e.g. EDI characteristics) and ask about caring situation.

Participants and settings: unpaid carers of people living with dementia across England. Both former (within five years[20]) and current carers will be eligible to take part, if they provid(ed) unpaid support to a relative, partner/spouse or friend with dementia. They will have to be at least 18 years old and able to provide written/oral consent. We will recruit participants through GP practices, memory services, pharmacies, social media (e.g., Twitter), previous studies, online dementia research websites (e.g. Join Dementia Research (JDR)), carer organisations and dementia services. We will also approach participants who took part in previous Stages if they agreed to be contacted.

EDI: similar considerations will apply as in previous Stages. We will contribute to carers replacement costs. If people do not wish to participate in a focus group e.g. due to language barrier an option will be provided to have an individual interview with an interpreter present.

Analysis: Two researchers will independently code the transcripts. The COM-B model and TDF will be used as a combined deductive framework for the analysis. The interview data will be managed using NVivo and SPSS will be used for descriptive data analysis of participant characteristics.

Outcome: understanding what influences behaviour through a behavioural diagnosis using the TDF and COM-B.

Dissemination: The results will be shared with participants. We will also disseminate the results through the Steering Committee and co-researchers will use their established links, e.g. Mobilise, Facebook groups, their own carers' groups, Dementia Friendly East Lothian, Local Advocacy groups etc. Co-researchers suggested following outputs to disseminate the findings: blog; leaflets; webinar; article for Alzheimer's Society journal or the UK Dementia Journal. We will also write peer-reviewed paper and co-present the results at a conference.

Stages 2c) Identifying intervention options; 36-43 months: research group discussions

Aim: to identify relevant intervention functions and policy categories following the COM-B and TDF analyses. As a first step we will present findings from the COM-B and TDF analyses (Stage 2b) at the Steering Committee meeting. We will discuss which themes should be prioritised, e.g. using the criteria of frequency (e.g. how many times does psychological capability feature as in influence in data we collected); elaboration (number of themes and sub-themes within each domain) and expressed importance (e.g. quotes from study participants expressing importance). Once we agree on key barriers and facilitators, together with research assistant and co-researchers, we will map these against nine intervention options and seven policy categories [4]. The intervention options include: education, persuasion, incentivisation, coercion, training, modelling, environmental restructuring,

restrictions, and enablement. Policy categories include: guidelines, environmental/social planning, communication/marketing, legislation, service provision, regulation and fiscal measures. The identified intervention options and policy categories will be assessed with co-researchers and with the Steering committee against APEASE criteria [4, 8]: acceptability, practicality, effectiveness, affordability, side-effects/safety and equity.

Stage 2d) Identifying content and implementation options; 36-43 months: research group discussions

Aim: to identify which techniques will deliver the intervention options and how will the intervention be put into practice.

Behaviour change techniques are the active ingredients within the intervention, designed to change the behaviour (e.g. goal setting, behavioural substitution, feedback on behaviour etc.). The behaviour change techniques will be selected using Behaviour Change Techniques Taxonomy App, which links taxonomy to Behaviour Change Wheel and in turn COM-B. It facilitates step-wise intervention design. The identified behaviour change techniques will be assessed with co-researchers and with the Steering committee against APEASE criteria [28, 32]. We will also identify behaviour change techniques mentioned within the focus groups in Stage 2b. The discussion with co-researchers, Steering committee and the findings from focus groups will then inform the development of intervention. We will also discuss the most appropriate style and mode of delivery of each technique. Examples of modes of delivery include professional conversation, posters, TV adverts, group counselling, peer support buddies, leaflets, website, mobile apps, legislation and pricing[4]. Decisions will be also made around who should deliver the intervention, and the intensity, frequency and duration of the intervention.

Outcome of Stages 2c and 2d: intervention will be developed

Dissemination: the range of dissemination strategies as mentioned in previous Stages. Key outputs will include conference, protocol published in peer reviewed journal describing the full detail of the intervention.

Stage 3: Feasibility study; 43-53 months

Aim: to test whether the proposed intervention is feasible and acceptable for unpaid carers.

Study design: feasibility study conducted in England. A quantitative (e.g., surveys) and qualitative (e.g. semi-structured interviews) mixed-methods design will be used to assess predefined progression criteria that relate to the evaluation design (e.g., reducing uncertainty around recruitment, data collection, retention, outcomes, and analysis) and the intervention itself (e.g., around optimal content and delivery, acceptability, adherence, likelihood of cost effectiveness, or capacity of providers to deliver the intervention).

Outcome: The end result will be a recommendation on whether an evaluation is feasible and acceptable.

Dissemination: the range of dissemination strategies as mentioned in previous Stages. Key outputs will include conference, lay summaries shared with participants and peer-reviewed publication.

Dissemination, Outputs and anticipated Impact

Project Management

I will be responsible for project co-ordination and management, budget management, ensuring effective communications within the team, overseeing our publication and dissemination strategy. I will also line manage the support staff through weekly face-to-face or online meetings.

Within this Fellowship application I have requested funding for support staff (research assistant) to the equivalent of 1 member of staff for a period of 3 years. The University of Kent will also match this support request. Depending on their career development needs and goals we will decide together with support staff what tasks they would like to be involved in/ lead on. Examples include support obtaining ethical and research governance approvals, preparing study documents, data analysis and leading on dissemination outputs. They will also have an opportunity to lead on work packages. As I am committed to supporting development of early career researchers I have also requested contribution of £3,000 towards training and development costs for the support post.

I will meet weekly with the support staff (either face-to-face or online) to oversee the project and discuss day to day study procedures and also identify together any training needs. They will be part of project management group and attend the Steering Committee.

Project Management Group (PMG) will meet monthly to ensure the project is on track and discuss project related issues. The group will include myself, support staff and eight co-researchers. Each member of the PMG will have an opportunity to chair these meetings. In addition to attending monthly meetings, co-researchers expressed interest in conducting the research itself including conducting individual interviews, focus groups, reviewing all study documents and support dissemination activities. We agreed with co-researchers that on top of monthly meetings they may contribute up to 10 hours per month to the research delivery. The type of activities and amount of commitment will be decided with each co-researcher depending on their availability at that time as their situation and health can fluctuate. Appropriate training will be provided.

The Steering Committee will meet online once a year at key stages of the project. The support staff, co-researchers and mentors will also attend all the Steering Committee meetings. The Steering Committee will include academics with relevant expertise, health and social care professionals, representatives from independent charities such as Age UK Canterbury, Carers Support West Sussex, Alzheimer's Society and commissioning officers.

The Brighton and Sussex Clinical Trials Unit (CTU): I will be supported by the Brighton and Sussex Clinical Trials Unit (CTU). We identified together that I will require support/ mentorship in three areas: data management, study management and statistical advice.

Mentors: As described in detail in section Research Support I will have regular meetings with my mentors to discuss and review my progress.

Finance: I will be responsible for overall management of the budget. The Finance team at University of Kent will govern the allocated budget and provide monthly statements of the income and expenditure.

Ethics

Before the start of the study, a favourable opinion will be sought from the Health Research Authority and/or University of Kent, for the study protocol, informed consent forms and other relevant documents e.g. advertisements. When needed Research Governance approvals will be also sought from participating local authorities and ADASS (Directors of Adult Social Care).

The proposed research will be compliant with the England's ethical requirements with regards to the collection, storage, processing and disclosure of personal information. In collaboration with Brighton & Sussex Clinical Trials Unit we will establish a Data Management Plan to ensure the study is conducted to high ethical standards.

All the data that will be collected will be stored securely. Name and contact details (if provided) will be stored separately to the other research data. They will only be used to contact participants about the research and to send them a summary of the findings. Paper documents will be kept in locked filing cabinets in locked offices. Electronic data will be password protected and stored securely on the University of Kent network drive.

The online survey will be conducted using Qualtrics software. The survey data will be protected by password and limited to the research team. Data stored on Qualtrics and transferred to the University of Kent will be encrypted.

All the interviews (individual interviews and focus groups) will be audio-recorded. Audio recordings will be made on encrypted recording devices. Files will be stored as audio files on the University network drive and the original recordings will be deleted from the recording device. Recorded audio files from the interview and focus groups will be sent to an external transcriber known to the University of Kent, which complies with the England's ethical requirements.

The University of Kent will be the Sponsor for this study based in Canterbury, Kent (UK).

Importantly each participant will receive an information sheet and will be asked to sign an informed consent (either online or printed) or provide verbal informed consent e.g. before each interview/focus group. Co-researchers (unpaid carers of people living with dementia)

will play a crucial role in development of these documents to ensure they are appropriate for participants.

There are several ethical challenges that may arise during the conduct of proposed work packages. For example, participants may find some questions difficult or upsetting to answer. It will be explained to participants that they do not have to answer any questions that they feel either unable or unwilling responding to. The information sheet will also include details of how to find/contact the carers' organisations, if the carer would like advice or support in relation to caring.

I have also experience with dealing with potential disclosures that someone might be at risk of harm.

Success Criteria

Please set out the measurements of success you intend to use and also the key risks to delivering this research and what contingencies you will put in place to deal with them. This section should identify appropriate actions that would reduce or eliminate each risk or its impact.

Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal (350 words)

The focus of this Fellowship is on unpaid carers of someone living with dementia. Therefore, unpaid carers played a central role in co-producing this Fellowship application. Over the period of twelve months (January 2023 until January 2024) we met together online on monthly basis for two hours. Throughout these meetings we get to know each other, we agreed how we wanted to work together, identified health and wellbeing concerns for unpaid carers, what the wellbeing and health-related goals carers would like to work towards etc. We also spoke about any barriers and facilitators when it comes to supporting carers to meet their own health and wellbeing goals. For example, it was highlighted that: 'It is important that professionals give advice that is purposeful and meaningful and would allow carers to act on it. Often advice around lifestyle changes is not applicable to carers.' In addition unpaid carers expressed that: 'There is no size fits all, the intervention has to be tailored to individual circumstances, it is important to speak to people and to listen to them.' In our meetings, we also spoke together about each work package. We specifically focused on what methods are the most appropriate way to collect the information, who should be the target population, how we would like to disseminate the results etc. They also provided feedback on the application itself to ensure that their voices run as golden thread through this application.

We were a team of ten people. Eight people self-identified as a former or current unpaid carer living across England and Scotland. Seven co-researchers will continue to work on the

Fellowship as co-researchers – this means they will be part of the core research team, they will own the research and be responsible for it. As one person passed away we will recruit a new co-researcher for this Fellowship. The co-researchers themselves represent a range of perspectives in terms of their experiences as carers. Importantly the group also included individuals with diverse characteristics: e.g. having English as a second language; being black, belonging to LGBTQ+ community and coming from range of socioeconomic backgrounds.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided (350 words)

Seven unpaid carers will continue to work on the Fellowship as co-researchers. As one person passed away we will recruit a new co-researcher for this Fellowship. We will specifically recruit someone from a minority background to further diversify the voices represented in our group.

The co-researchers will participate in monthly two hours online project management meetings. The frequency, format and length of meetings were suggested by co-researchers themselves and this worked really well for us in the last twelve months. If the person cannot attend the meeting, they will be offered a replacement meeting – a practice that we established while working together on this application. Each person will receive ahead of the meeting study packages (either printed or electronic version) including any materials that will be discussed during the meeting. Co-researchers will have an opportunity to chair these meetings and facilitate any breakout groups during the meetings. They will be paid for their attendance at the meetings including contribution to their internet costs. Importantly to allow them to contribute we also budgeted for carer replacement costs: some carers of people living with dementia may require to have someone in place instead of them to look after a person living with dementia while working on the project. Co-researchers will also attend all the Steering Committee meetings.

In addition to attending monthly meetings, co-researchers expressed interest in conducting the research itself including conducting individual interviews, focus groups, reviewing all study documents and support dissemination activities. We agreed with co-researchers that on top of monthly meetings they may contribute up to 10 hours per month to the research delivery. The type of activities and amount of commitment will be decided with each co-researcher depending on their availability at that time as their situation and health can fluctuate.

Co-researchers will be provided with training to supplement and enhance their existing skills so they are able to carry out research activities related to the project. Each of them will be also able to attend two conferences throughout Fellowship.

Training & Development and Research Support

Proposed training and development programme (1000 words)

1. **Training focused on behaviour change interventions:** The proposed research is informed by Behaviour Change Wheel model[4]. In 2022-2023 I attended two courses in Behaviour Change Wheel Model: Behaviour Change Interventions: Introductory Principles and Practice (introductory) and Applied Principles and Practice of Behaviour Change (intermediate) to support the development of this application. During the Fellowship I would like to gain more in depth-knowledge of Behaviour Change Wheel model and therefore I propose to attend two Advanced Modules in Behaviour Change: Principles and Practice of Behavioural Systems Mapping and the Behaviour Change Intervention Ontology (BCIO): How to use the BCIO for different research and applied purposes. I plan to attend these courses during the first year of the Fellowship. These courses are delivered by the developers of the model based at Centre for Behaviour Change, UCL. In addition to gaining a specific knowledge, attending these courses will also allow me to connect to other researchers interested in this specific approach.

   Importantly, Behaviour Change Wheel model represents one of the approaches to behaviour change. There are other approaches to developing interventions e. g. Intervention Mapping[7] which provides a framework for developing theory-based and evidence-based interventions applied to health promotion and disease prevention. To be able to support other researchers in the future that aim to improve the wellbeing and health of unpaid carers of people with dementia, I also propose to attend 5-day summer course on Intervention Mapping: Designing Theory-based and Evidence-based Programs at Maastricht University in the Netherlands. Being trained in two key approaches to behaviour change intervention development will enable me to work beyond this Fellowship on a range of projects and connect with people who apply these approaches in their work. I plan to attend this course in second year of the Fellowship.

2. **Training in Co-production**: as a researcher I am committed to work in co-produced way together with unpaid carers of people living with dementia, other academics, health and social care professionals. While co-producing this application over the period of twelve months together with unpaid carers I have identified two key areas where I need additional training: 1. working with conflict and confrontation in teams; and 2. facilitation skills. As such I plan to attend three courses delivered by Gestalt Centre in London during the first year of Fellowship: Group Facilitation: Key concepts, skills and personal development (one day in person workshop); Working with Conflict and Confrontation in Groups and Teams (two days in person workshop);

and Group Facilitation Training Intensive (nine days in total spread over four months).

3. **Equality, Diversity and Inclusion (EDI)**: As a researcher, I am aware that the research I conduct mainly reflects experiences of those who self-identify as e.g. white, British, heterosexual and from middle class. As such in this Fellowship, I ensured that we will take a proactive steps to recruit people from diverse backgrounds – this includes co-researchers, members of the Steering Committee and research participants. While as an EDI representative at University of Kent I have opportunities to attend range of training sessions, in this Fellowship I would like to specifically attend training sessions related to EDI issues when conducting research. Therefore, I plan to attend following training throughout the Fellowship: a) Improving the inclusion of under-served groups in health research (online webinar by INVOLVE); and b) An Ethical Perspective on Implementing Equal Opportunities in PPI (online webinar).

   Importantly as part of NIHR Applied Research Collaboration I am member of a working group of researchers interested in inclusion of individuals living with dementia and their carers in research.

4. **Impact and dissemination training:** I aim to become a researcher who has an influence beyond academia. Specifically, I would like to make the needs of unpaid carers visible to present and future health and social care professionals as well as policy makers. I plan to attend following courses: a) Impact Training for Researchers (online course provided by Fast Track Impact over 5 weeks; first year of the Fellowship) and Compelling Communication Skills (6 weeks course delivered by University of Cambridge; second year of the Fellowship).

   In addition I will apply for The Royal Society Pairing Scheme. Each year 30 research scientists are paired with UK parliamentarians and civil servants. They learn about each other's work by spending time together in Westminster and the researcher's institutions. Those taking part gain an insight into how research findings can help inform policy making, and come away with a better understanding of how they can get involved.

5. **Leadership training:** I would describe a good leader in three words: kind; adaptive and self-aware. I had an opportunity to attend the NIHR Virtual Leadership Programme for social care researchers which was a scaled down version of NIHR Future-Focused Leadership Programme. As the content of that course reflected my

own values I would like to attend the NIHR Future-Focused Leadership Programme as part of this Fellowship.

6. **Software skills:** I would like to become an expert on two softwares that I regularly use in my research and will be also central to delivery of this fellowship: NVivo and Qualtrics. As such I suggest to attend two courses: NVivo Expert Certification: this will enable me to be an in-house NVivo expert and advisor for other researchers across University of Kent. I will also do Qualtrics online courses, again allowing me to raise my profile in the research community.

7. **Specific skills and knowledge needed for the next stage of the project (after Fellowship):**
   - Implementation Science Training: designed to build capacity and prepare for implementation in particular settings; third year of the Fellowship
   - Economic Evaluation Online: An introductory course; third year of the Fellowship

8. **Conferences:** Conferences are incredible opportunity to present own research, to learn about up to date evidence and make important connections. As such I plan to attend at least one conference between years 2-4 of the Fellowship. As such I will focus on conferences in three areas: dementia; health psychology/behaviour change; and ageing.

Research Support and supervisor

Dr Rosie Ashworth
NHS Scotland Neuroprogressive and Dementia Network
The Lead for the Patient and Public Involvement group

Description of support provided 250 words

Rosie leads a group known as 'Partners in Research' that includes people living with dementia and their unpaid carers. They work together on a range of activities including book writing, developing research proposals, reviewing research materials and raising awareness. Rosie is also a Research Fellow at the University of Edinburgh with a special interest in co-produced research.

Rosie played a central role in development of this Fellowship application as she provided me with an advice on co-production in last twelve months. She also helped me to co-facilitate majority of the meetings with unpaid carers in last twelve months as we worked together on this application. We have worked together on co-producing a blog, academic paper, podcast episode and another funding application.

Rosie will continue to support me regularly throughout the Fellowship. She will provide an advice on any issues with co-production, co-facilitate meetings with unpaid carers when needed and she will be part of the Steering Committee. We will have monthly online meetings to discuss the progress of this Fellowship. She will be also involved in key research outputs from this Fellowship.

Importantly, Rosie has established links with other researchers in dementia working in co-produced way and will help me to network with other academics working in similar area. We plan to establish a peer support group for researchers in dementia who use co-production. In addition we plan to co-produce together with unpaid carers of people living with dementia a training course on co-production for researchers working in dementia.

Dr Sarah Hotham

Centre for Health Services Studies, University of Kent

Senior Research Fellow

Description of support provided 250 words

Sarah is a Senior Research Fellow and BPS Chartered Psychologist. She is an expert in behavioural health psychology, with a specific focus on areas of nutrition (breastfeeding, obesity and malnutrition), sanitation and training of healthcare professionals. Her current and previous research has focused on three UN Sustainable Development Goals: Zero hunger; Good health and wellbeing (i.e. decrease the risk of non-communicable diseases); and Clean water and sanitation.

Sarah has an extensive expertise in supervising projects with a focus on behavioural science approaches to design, delivery and evaluation of health behaviour change interventions. She is also interested in Implementation Science approaches to evaluation and how evidence can be translated into policy and practice.

Sarah will provide me with expertise during the Fellowship on the design of the health behaviour change intervention as well as the feasibility study. Importantly, Implementation Science is an area where I lack an experience and expertise and this is the area where I would like to gain more knowledge. Sarah will support me with this. We will meet regularly once per two months to discuss the progress of the Fellowship. She will also be involved in the outputs and she will be member of the Steering Committee.

Importantly Sarah will also support me to think about my career beyond this Fellowship. She is based at the same University as I am and she will mentor me in regards to advancing my own career at University of Kent.

Professor Naji Tabet

Brighton & Sussex Medical School

Professor, Dementia and Old Age Psychiatry

Description of support provided 250 words

Naji is the Director of the Centre for Dementia Studies (CDS) and the Course Leader of the MSc Dementia Studies at Brighton and Sussex Medical School. He also leads the Dementia Theme at the NIHR Applied Research Collaboration KSS. He is also the Dementia Speciality Co-Lead for NIHR Clinical Research Network KSS.

Naji has led the unit on several important phase II-IV clinical trials trying to advance the treatment of Alzheimer's disease. Importantly he is also involved in studies that investigate the role of non-pharmacological interventions, lifestyles and co-morbid physical illnesses in disease progression. He has been investigating the role of antioxidants, dietary and endogenous, and mechanistic metabolites in Alzheimer's disease and their interaction with currently available treatments. In addition to his research activities, clinically, he holds an Honorary Consultant post at Sussex Partnership NHS Foundation Trust where he delivers a diagnostic and treatment Memory Assessment Service to Trust patients with memory complaints.

Naji will support me throughout the Fellowship and we will meet once per three months to discuss the progress of the Fellowship. He will also help me to think about next steps beyond the Fellowship. He will be involved in the outputs and be member of the Steering Committee.

As Naji is a practicing clinician and therefore I will use our conversations to better understand how people with dementia and their unpaid carers are supported in the NHS.

Professor Lindsay Forbes

Centre for Health Services Studies at the University (CHSS) of Kent

Clinical Professor of Public Health

Description of support provided 250 words

Lindsay is the Theme Lead for Public Health at the NIHR Applied Research Collaboration Kent, Surrey and Sussex. She is also Clinical Professor of Public Health at the Centre for Health Services Studies at the University (CHSS) of Kent.

Lindsey is a public health scientist with interests in understanding patterns of ill health and health systems to address the needs of the population. She is a public health specialty lead for Kent Surrey and Sussex NIHR Clinical Research Network and an adviser for the NIHR Research Design Service South East. Lindsey is on the General Medical Council specialist register in public health medicine. Most recently her work has been in two main areas: design and evaluation of primary care services, and systems to build evidence of what works to improve the health of the public. She started off her academic public health career in epidemiology, although over the years she has developed a much broader set of research skills, including complex intervention design and evaluation.

Lindsey has well established links across both health and adult social care. She will input into study design and advice specifically on Stage 1 (Needs assessment) as well as developing of the intervention. Lindsey will especially mentor me to think about next stages of the proposed research after the Fellowship (evaluation and implementation). She will be involved in

supporting dissemination activities - e.g. making them relevant for public health specialists across local authorities. She will be part of the Steering Committee.

Collaborations (600 words)

I plan to undertake the award at Centre for Health Services Studies (CHSS). CHSS is a centre of research excellence which undertakes high quality research into a wide range of health and social care systems and health and care services issues at local, national and international levels. CHSS builds the evidence base for policy and practice to prevent ill-health, promote well-being and meet needs for care, focusing on primary and community services, public health and social care working with academic collaborators across the UK and the world. CHSS works with local government and health services across Kent, Surrey and Sussex to build research capacity and culture and address the key health and social care research needs across the region.

CHSS has a substantial number of links with other organisations including National Institute of Health Research (NIHR) Applied Research Collaboration Kent, Surrey and Sussex (ARC KSS) and the Kent Adult Social Care Partnership. ARC KSS is the national lead for social care. Primary and Community Care and Living Well with Dementia are also key themes of the ARC, with active and interlinked research programmes. There are well-established relationships between the ARC theme leads (primary care lead is Professor Harm van Marwijk in Brighton and Sussex Medical School and Professor Naji Tabet (my mentor) is theme lead for dementia). As an employee of University of Kent I have an access to senior researchers who are part of ARC and introductory emails will be send by Professor Lindsay Forbes (my mentor) and/or Professor Naji Tabet. In addition, I will draw on network of local contacts in public health and also primary care of Professor Lindsey Forbes who is a Clinical Professor of Public Health at Centre for Health Services Studies, University of Kent and a Co-Lead for Public Health Practitioner Communities and Networks at the NIHR Applied Research Collaboration Kent, Surrey and Sussex. Furthermore, I will join the Society of Academic Primary Care (SAPC) which would connect me to all the country's leading researchers in primary care.

Kent Adult Social Care Partnership led by Professor Ann-Marie Towers (University of Kent) supports the social care sector in Kent by conducting high quality research relating to local social care priorities. One of the key part of Kent Adult Social Care Partnership are Communities of Practice which bring together people who have different expertise but a shared goal. Professor Towers is my line manager and she will support me in accessing links across adult social care.

Importantly being able to be based at Kent is also very strategic as Kent, Surrey and Sussex are in an ideal position to lead on ageing and living well with dementia research. As a region, we have a higher proportion of older people, and people living with dementia and our ageing population is growing faster than the national average. While working at University of Kent, I

have formed relationships with NIHR Clinical Research Network Kent, Surrey and Sussex and independent charities such as Age UK Canterbury, Carers Support West Sussex, Alzheimer's Society (local branches) and commissioning officers based locally. I would continue to work closely with them during the award and seek their expertise and advice.

Lastly, I will be supported by the Brighton and Sussex Clinical Trials Unit (CTU) that provides support and expertise for clinical researchers in the South East in the development of funding applications, trial design, study management, quality assurance, data analysis, dissemination and trial close-down. We identified together that I will require support/ mentorship in three areas: data management, study management and statistical advice.

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